

Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

dBs and Link Budgets

Dr Toby Whitley



UNIVERSITY OF
PLYMOUTH



dB calculations

- Allows us to deal with large and small numbers easily.
 - dBs compress the dynamic range and put everything on the same scale
 - For example:

Distance	dB
1mm	-30dB
1m	0dB
10m	10dB
100m	20dB
1km	30dB
10km	40dB

dB Calculations

- It's all about ratios
 - Difference between 100m and 1km :
 - $10\log(100/1000) = -10\text{dB}$
 - Half the signal: $10\log(1/2) = -3\text{dB}$
 - Double the signal: $10\log(2/1) = 3\text{dB}$
- It's all about reference points
 - What is the first table reference to?
- In RF and comms typically milliwatts and watts.

Typical receive power examples

Communications system	Received power
GSM	-51dBm
GPS	-155dBW (-125dBm)
WIFI	-75dBm or better
Voyager Space Ships	-231dBm

Transmit powers

- Voyagers:
 - 18 watts @ 8GHz
- GSM
 - Varies @ 900 & 1800MHz
- Wifi
 - 100mW (1Watt) @2.4GHz
- GPS
 - 25.6 Watts @1573MHz

dB Calculations

GSM 900:

Power level number	Power output level dBm
2	39
3	37
4	35
5	33
6	31
7	29
8	27
9	25

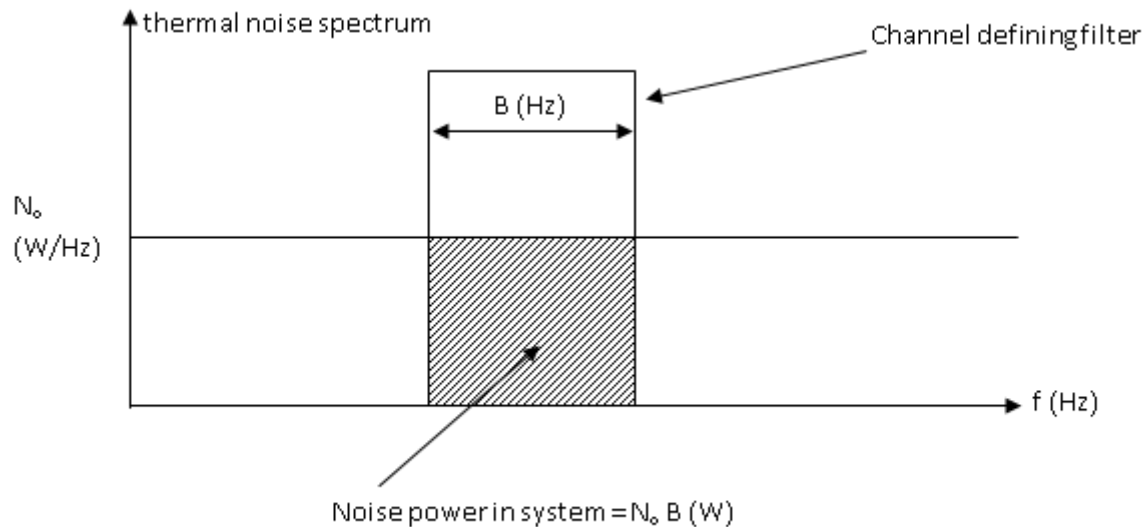
Down to 19 at 5dBm

Why 5 bars for phone signal strength?

GSM reception levels for voice calls

- -105 to -100 = Bad/drop call
- -99 to -90 = Getting bad/signal may break up
- -89 to -80 = OK/shouldn't have problems, but maybe
- -79 to -65 = Good
- Over -65 = Excellent

Thermal Noise and Noise Figure



Thermal noise power (N_o) is distributed evenly across a very wide spectrum and is measured in watts per Hertz (watts/Hz). N_o is given by the product of Boltzmann's constant and absolute temperature ($N_o = kT$).

Thermal Noise and Noise Figure

- The thermal noise power is then given by the product of N_0 and B , where N_0 increases in proportion to temperature.

Boltzman's Constant = $K = 1.3806488(13) \times 10^{-23}$ joules/K

- Converting to dB, at room temperature ($20^\circ\text{C}=293\text{K}$),
- $N_0 = -174 \text{ dBm/Hz}$ or -204 dBW/Hz .

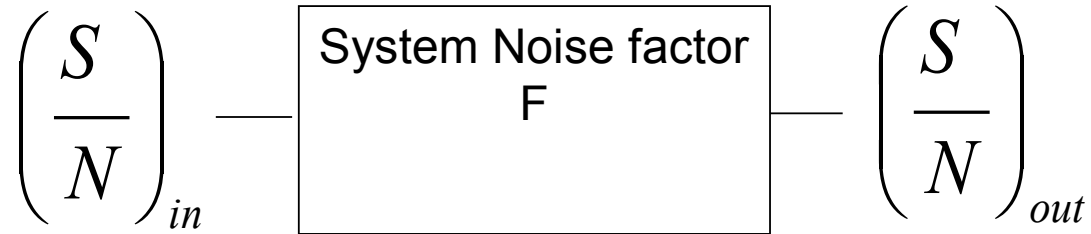
Thermal Noise

Then if a receiver is perfect apart from thermal noise, the total noise power with which the signal must compete referred to the input of the receiver is:

$$N = -204 + 10 \log_{10}(B) \quad (dBW)$$

$$N = -174 + 10 \log_{10}(B) \quad (dBm)$$

Noise Factor



However, usually the receiver also contributes some noise over and above the thermal noise. The amount of extra noise depends upon the electronics used and is described by the **noise factor** or **noise figure** (when measured in dB).

Noise Factor, Figure

$$\text{noise factor } F = \frac{\left(\frac{S}{N}\right)_{in}}{\left(\frac{S}{N}\right)_{out}}$$

$$\text{or in dB, noise figure } F(\text{dB}) = \left(\frac{S}{N}\right)_{in} (\text{dB}) - \left(\frac{S}{N}\right)_{out} (\text{dB})$$

$$N = -204 + F(\text{dB}) + 10\log_{10}(B) \quad (\text{dBW})$$

Path Loss

$$L = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \text{ dB}$$

Where R is the distance from the transmitter and λ is the wavelength of the signal given by the speed of light over frequency $\sim c/f$

This gives a positive value for loss, you may see it written with λ on top and $4\pi R$ on the bottom which gives a negative result.

Path Loss

$$L = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \text{ dB}$$

Where R is the distance from the transmitter and λ is the wavelength of the signal give by the speed of light over frequency $\sim c/f$

This gives a positive value for loss, you may see it written with λ on top and $4\pi R$ on the bottom which gives a negative result.

Plympton 11 km
 P_{Tx} 400W

Caradon 19 km
100 kW

$$\left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} L = 20 \log_{10} \left(\frac{4\pi R}{\lambda} \right) \\ f = \underline{800 \text{ MHz}} \end{array}$$

WIFI IN MY GARAGE ASSUME 2.4 GHz 20 MHz BW $P_t = 100 \text{ mW}$

Distance 25m

P_r & SNR